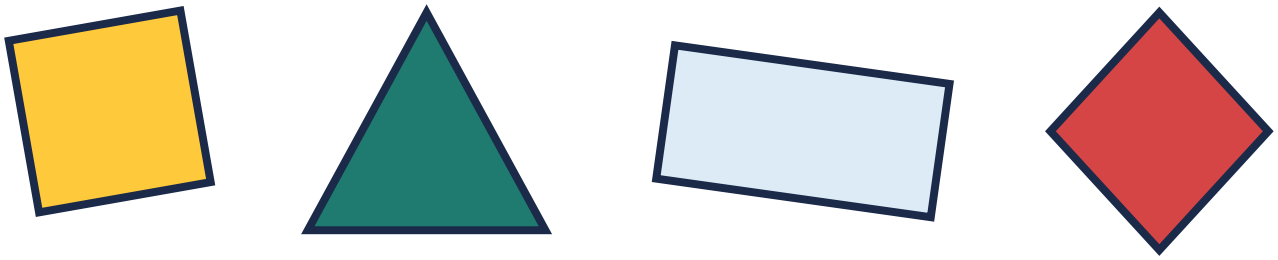
FOR PARENTS • PRIMARY 5 HOLIDAY PROGRAM

Weekly Dose of Thinking Math

One dose a week for five weeks: short sets of questions for every Math Inquirers lesson, a tip for you, and the answers at the back.



Notice it. Test it. Explain why.

Math Inquirers Holiday Program
Preciouslyours Pte Ltd

BEFORE YOU START

00 FOR GROWN-UPS How to help

1. **Ask before you tell.** Start with "What do you notice?" and let your child answer first.
2. **Use real things.** Coins, notes, cups, string, paper strips or blocks. Try it with objects or a drawing before writing.
3. **Give hints in three steps.** Look again. Try it with objects. Try a smaller example. Say the answer last.
4. **Ask "How do you know?"** even when the answer is right. The reason matters as much as the answer.
5. **Praise the thinking.** "I like how you checked that" works better than "You are clever".
6. **Keep it short.** Ten minutes and two or three questions is enough. Stop while it is still fun.
7. **Answers are at the back.** Look only after your child has tried.

> **Hard question? Great. Getting stuck is where the thinking happens.**

One dose a week

Each week has two lessons and about 18 questions. Do two or three a day, Monday to Friday, and keep the weekend for the Go further questions if your child enjoys them.

The three levels

Start here is for everyone and can be tried before the topic is taught. **Think harder** uses the idea from the lesson in a new setting. **Go further** and the **Investigation** are for when your child wants more, and it is fine to leave them for another day.

What your child is doing in class

Ten 90-minute live online lessons over five weeks, in small groups. Each lesson starts with a hands-on puzzle, then a short teaching part, then the workbook. These pages practise the same ideas at home.

W1 Million Puzzles

LESSON 1 • NUMBERS TO 10 MILLION

Start here

1 Write 4 030 205 in words.

2 What is the value of the digit 6 in 3 621 480?

3 Round 5 482 716 to the nearest thousand.

4 Use seven digit cards, 0 1 2 3 4 5 6, each once. Make the greatest 7-digit number you can, and then the smallest. (A 7-digit number cannot start with 0.)

Think harder

5 How many 7-digit numbers have digits that add up to 2? List them all.

6 Put ONE pair of brackets into 2 + 3 × 4 + 5 to make 25. Then put one pair in a different place to make 29.

What your child learns

Reading, comparing and rounding numbers up to 10 million, and riddles about large numbers.

Tip for you

Find a big number in the news, such as a population or a budget. Ask your child to say it in words and round it to the nearest million.

Ask

"Which digit decides how you round?"

7 A 7-digit number rounded to the nearest thousand is 2 500 000. Its digits add up to 9. Write the range first (the smallest and the greatest number that round to 2 500 000). Then find ALL the numbers that fit. How do you know you have them all?

Go further

8 Make up your own digit riddle for a 7-digit number. Use at least 5 clues (for example: all digits different, a rounding clue, a clue that compares two digits, a digit sum). Only ONE number may fit. Swap with a partner: do you get the same answer? How can you be sure your riddle has only one answer?

9 Write 1 000 000 as the product of two whole numbers that have no zero digits at all.

W1 Fraction Feats

LESSON 2 • FRACTIONS: FOUR OPERATIONS

Start here

1 4 friends share 3 chocolate bars equally. Draw the bars and cut them. How much chocolate does each friend get?

2 Siti ate 1/2 of a pizza and Ben ate 1/4 of the same pizza. Draw the pizza. What fraction did they eat altogether? What fraction is left?

3 A pizza is cut into 12 equal slices (draw it). Ravi eats 1/3 of the pizza and Mei eats 1/4 of the pizza. How many slices did they eat? What fraction of the pizza is left?

4 Find 3/5 of 15 sweets. (Draw 15 dots and put them into 5 equal groups.)

Think harder

5 Write 3/8 as the sum of two different unit fractions in TWO different ways. How do you know when to stop looking?

6 Ben ate 1/3 of a pizza. Ali ate 1/4 of what was left. What fraction of the pizza is left?

What your child learns

Adding, subtracting, multiplying and dividing fractions, using bar models and pictures.

Tip for you

Share a block of chocolate or a cake. Ask what is left after you eat 1/3 and then half of what remains.

Ask

"Can you show it with a bar?"

7 1/2 − 1/3 = 1/6 and 1/3 − 1/4 = 1/12. Use the pattern to work out 1/4 − 1/5 and 1/9 − 1/10.

Go further

8 Write 1 as the sum of three different unit fractions. Find all the ways.

9 Work out 1/2 × 2/3 × 3/4 × 4/5 × ... × 99/100.

W2 Before and After

LESSON 3 • FRACTION WORD PROBLEMS

Start here

1 Ali has 18 counters and Mei has 6 counters. Ali gives Mei 4 counters. How many counters does each child have now? What is the total before, and what is the total after? (Use counters or draw dots.)

2 Ben is 10 years old and his father is 38. How old will each of them be in 5 years? How many years older than Ben is his father now? And in 5 years?

3 Siti had $60. She spent 1/4 of it on a book and 1/3 of the remainder on a T-shirt. How much money did she have left?

4 Ali had 26 stamps and Ravi had 34. How many stamps must Ravi give Ali so that they have the same number?

Think harder

5 Ravi spent 1/4 of his money on Monday and 2/5 of the remainder on Tuesday. He spent $66 altogether. How much money did he have at first?

What your child learns

Fraction word problems: before-and-after situations and 'fraction of the remainder', solved with bar models.

Tip for you

Make up a story: "I spent 1/4 of my money, then 1/2 of the rest, and had $12 left." Let your child draw it and work backwards.

Ask

"What is the whole here, and what is it now?"

6 A box had red beads and blue beads. 3/7 of the beads were red. Then 30 red beads were added. Now 3/5 of the beads are red. How many beads were in the box at first? How many are in the box now?

7 Mrs Tan is 41 years old and her son is 9. How many years ago was her son's age 1/5 of her age?

Go further

8 Make up your own before-and-after story in which the total stays the same and a fraction is used in the After (like "Ali has 1/2 as many as Ben"). Solve it with a model, swap with a partner and check each other. What if the fraction in your story were changed?

9 Ali has 18 counters and Mei has 6. Ali may give Mei some counters (he can give none, or up to all 18). After the giving, one child has exactly 2, 3, 4 ... times as many counters as the other (the two numbers are not equal). List ALL the ways. How do you know you have them all?

W2 Triangle Areas

LESSON 4 • AREA OF TRIANGLE

Start here

1 On 1-cm grid paper draw a rectangle 6 cm by 4 cm. Cut it along a diagonal into two triangles. What is the area of each triangle? (Fold or cut to check that the two triangles are the same.)

2 A square has sides of 8 cm. Draw it. Put a point anywhere on the top side and join it to the two bottom corners. Cut the three pieces out: do the two side pieces fit exactly on the middle triangle? What is the area of the middle triangle?

3 On grid paper draw a right-angled triangle with a base of 4 cm and a height of 3 cm. Draw the same triangle again, turned upside down, so that the two triangles make a rectangle. What is the area of ONE triangle?

4 Draw a rectangle 5 cm by 4 cm. Mark a point on the top side. Join it to the two bottom corners. What is the area of the middle triangle? Try a different point: does the area change?

Think harder

5 A triangle has an area of 45 cm². Its base is 10 cm. What is its height?

What your child learns

Area of a triangle: base and height, why the area is half of a rectangle, and triangles in composite figures.

Tip for you

Cut a rectangle of paper in half along the diagonal. Ask why each triangle is half of the rectangle.

Ask

"Which side is the base, and where is the height?"

6 A right-angled triangle has sides of 12 cm, 16 cm and 20 cm. What is its area?

7 A triangle and a square have the same area. The square has sides of 6 cm. The triangle has a base of 9 cm. What is the height of the triangle?

Go further

8 Make up your own shaded-area puzzle. Draw a rectangle with the lengths of its sides, mark points on the sides and join them to make a shaded triangle and two or three white triangles. Work out the shaded area in TWO ways (split it, or take the white triangles away) and check that they agree. Swap puzzles with a partner.

9 A rectangle is 12 cm by 8 cm. P is the middle of the top side and Q is the middle of the right side. What is the area of the triangle joining P, Q and the bottom-left corner?

W3 LESSON 5 • VOLUME Volume Vault

Start here

1 Build a box 3 cubes long, 2 cubes wide and 2 cubes high with 1-cm cubes (or draw it layer by layer). How many cubes are in the bottom layer? How many layers are there? How many cubes are used altogether?

2 Use 12 cubes to build a flat rectangle, one cube high. How many different rectangles can you make? (Turning a rectangle round does not make a new one.) Write the length and width of each.

3 A tank has a base of 10 cm by 10 cm. If the water is 1 cm deep, there are 100 one-cm cubes of water. How many cubes of water are there if the water is 3 cm deep? 10 cm deep? How many litres is the water when it is 10 cm deep? (1 ℓ = 1000 cm³.)

4 How many 1-cm cubes (dice) fill a box 5 cm long, 4 cm wide and 3 cm high? Use dice, or draw one layer and count the layers.

Think harder

5 How many cubes with 2-cm edges can be packed in neat layers into a box 9 cm long, 6 cm wide and 5 cm high?

What your child learns

Volume of cuboids, water in tanks, stones that sink, and painted-cube puzzles.

Tip for you

Fill a box with sugar cubes or blocks. Ask for the number of cubes before counting, then check.

Ask

"How many cubes in one layer, and how many layers?"

6 A tank is 1/3 full of water. After 12 ℓ more is poured in, it is 3/5 full. How many litres does the tank hold when full?

7 A cuboid has a square base and a height of 5 cm. Its volume is 245 cm^3. How long is each side of the base?

Go further

8 A big cube is painted on the outside and cut into small cubes of the same size. 96 small cubes have exactly one painted face. How many small cubes are there altogether?

9 How many different cuboids can be built with exactly 64 one-centimetre cubes, using all of them each time?

W3 LESSON 6 • RATE Rate Racers

Start here

1 4 apples cost $2. How much do 8 apples cost? How much do 12 apples cost? How much does 1 apple cost? (Make a table if it helps.)

2 A printer prints 24 pages a minute. How long does it take to print 360 pages?

3 Pack A has 3 pens for $2. Pack B has 6 pens for $5. Which is the better buy? (Hint: how much would 6 pens cost if you bought Pack A twice?)

4 A tap fills a 120 ℓ tank at 8 ℓ a minute. How long does it take?

Think harder

5 A tap fills a tank in 12 minutes at 15 ℓ a minute. When a second tap is also turned on, the tank fills in 9 minutes. How many litres a minute does the second tap fill?

6 Eggs are sold in boxes of 10 for $3.30 and boxes of 6 for $2.10. Siti needs at least 28 eggs. What is the least she can pay?

What your child learns

Rate: the amount for one, better buys, taps filling a tank, and working out a charge 'or part of' an hour.

Tip for you

At the supermarket, compare two pack sizes of the same item. Which costs less for one?

Ask

"What does it cost for just one?"

7 A car park charges $1.50 for the first hour and $1 for every half hour or part of a half hour after that. How much does it cost to park from 10.15 a.m. to 2.50 p.m.?

Go further

8 Printer P prints a job in 6 minutes. Printer Q prints the same job in 3 minutes. How long do they take working together?

9 Siti buys pens at 3 for $2 and sells them at 2 for $2. How many pens must she buy and sell to make a profit of $10?

W4 Percentage Puzzles

LESSON 7 • PERCENTAGE

Start here

1 A hundred-square has 100 small squares. 35 of them are shaded. What percentage is shaded? What percentage is NOT shaded? (Percent means “out of 100”.)

2 Percent means “out of 100”, so 50% is a half. Find 50% of $80. Then use it to find 25% of $80 (half of 50%) and 10% of $80 (one tenth).

3 A book costs $50. It has 10% off, which is one tenth of the price. How much money is taken off? What do you pay?

4 24 of the 40 pupils in a class walk to school. What percentage walk to school?

Think harder

5 At a hawker centre, a plate of chicken rice went up from $4 to $5. A month later, it went back down from $5 to $4. (a) Find the percentage increase from $4 to $5. (b) Find the percentage decrease from $5 to $4. (c) Both changes are $1. Why are the two percentages different? Hint: percentage change = the change out of the amount you START with.

What your child learns

Percentage as a fraction of 100, finding a part, discount, GST and percentage change.

Tip for you

Look at a sale tag. Ask for the price after 20% off, then check with a calculator.

Ask

"Percentage of what? What is the 100%?"

6 A box of cereal costs $15 in two shops. Shop X: "Buy 3, get 1 free!" Shop Y: "30% off every box!" Mei buys 4 boxes. (a) How much does she pay at each shop? Which is the better buy, and by how much? (b) Shop X's deal is really how many percent off?

7 Siti says, "15% of $80 is $8 + $4 = $12." Explain her method. Use it to find 15% of $46.

Go further

8 Make up your own percentage trap: a story in which a price goes up by a percentage and then down by a percentage (or two percentages are added) and someone gets the wrong answer. Use real numbers to show what really happens. Swap with a partner: can they find the mistake? What if the two percentages were different?

9 After a 20% discount, a shirt costs $40. Ben says the price before the discount was $48. Is he right? Find the price before the discount.

W4 Angle Chasers

LESSON 8 • ANGLES, TRIANGLES AND QUADRILATERALS

Start here

1 Draw any triangle on paper and tear off its three corners. Put the three corners side by side with their points touching. What do you notice? What do the three angles add up to? (A straight line is a half-turn, 180°.)

2 Two angles sit side by side on a straight line. A straight line is 180°. One of the angles is 127°. What is the other angle?

3 A round pizza is cut into 8 equal slices. A full turn is 360°. What is the angle of one slice at the centre? What is the angle of 2 slices together?

4 Two angles of a triangle are 48° and 67°. Draw the triangle with a ruler and a protractor and measure the third angle. What do all three angles add up to?

Think harder

5 In parallelogram ABCD, angle A is 3 times angle B. Find angle A and angle B. Give your reason.

6 An isosceles triangle has an angle of 110°. Find the other two angles. Why is there only one answer?

What your child learns

Angle rules for triangles and quadrilaterals, finding unknown angles and giving a reason for each step.

Tip for you

Tear off the three corners of a paper triangle and put them side by side. What do they make?

Ask

"Which angle rule did you use, and why?"

7 ABCD is a parallelogram and E is a point on DC with DA = DE. Angle ABC = 56°. (The figure is the same one as in Start here Q4.) Find angle AEC and angle EAB. Write the reason for each step.

Go further

8 Two identical right-angled triangles with angles 30°, 60° and 90° are joined with their shorter sides (the sides opposite the 30° angles) touching, with the right angles together, to make one big triangle. What are the angles of the big triangle?

9 Make up your own angle-chase puzzle: draw a figure with a triangle and a parallelogram (or a rhombus) sharing a side, give just two or three angles, and ask for an angle that needs at least THREE steps. Write the reason for every step in your answer. Swap with a partner and check each other. What if you changed one of the given angles?

W5 Pattern Generalisers

LESSON 9 • PATTERNS AND NUMBER SENSE

Start here

1 5, 9, 13, 17, … What is the 10th number in this pattern?

2 Matchstick squares in a row use 4, 7, 10, … sticks. How many sticks are needed for 8 squares?

3 2, 6, 10, 14, … Is 50 in this pattern? Explain.

4 There are 15 bicycles and tricycles in a shed. They have 37 wheels altogether. How many tricycles are there?

Think harder

5 Beads repeat in the order red, red, blue, yellow, yellow, yellow. What colour is the 50th bead? How many yellow beads are in the first 100?

6 40 square tables in one long row seat $2 \times 40 + 2 = 82$ people. How many can sit if the tables are put in 8 rows of 5 tables instead?

What your child learns

Patterns and number sense: finding the rule, predicting far-away terms, and checking with a table.

Tip for you

Make a matchstick pattern. Ask for the number of sticks in the 20th figure before you build it. Week 5 note: lessons 9 and 10 are four days apart (Wed 23 and Sun 27 December), so there is time for the take-home parts in between.

Ask

"Can you test your rule on the next figure?"

7 A farm has chickens and goats. There are 35 heads and 94 legs. How many goats are there?

Go further

8 A staircase is built from squares: step 1 is 1 square, step 2 adds a column of 2 (3 squares altogether), step 3 adds a column of 3 (6 squares). How many squares are there in the staircase with 10 steps?

9 Make up your own growing pattern with matchsticks, dots or tiles. Draw the first three figures, work out the rule in words, and write two questions for a partner: the number of sticks in figure 50, and which figure uses exactly 100 sticks (or explain why no figure does). Swap and check. What if the pattern started with a different first figure?

W5 LESSON 10 • MIXED STRATEGIES Challenge Day

Start here

1 A number is multiplied by 4, then 6 is taken away, and the answer is 50. What was the number?

2 Ali has 1/2 of his money left after buying a book for $9. How much did he have at first?

3 Two numbers add up to 40 and the bigger is 8 more than the smaller. What are the two numbers?

4 Ravi, Siti and Ben each have a different pet: a rabbit, a hamster and a bird. Ravi has no bird and no hamster. Siti has no hamster. Who has the bird?

Think harder

5 Siti spent 2/5 of her money, then 1/3 of what was left. She had $24 left at the end. How much did she start with?

6 How many 3-digit numbers have digits that add up to 5? List them all in order.

What your child learns

Choosing a strategy: guess and check, working backwards, making a table and drawing a model.

Tip for you

When your child is stuck, ask "Can you draw it?" or "Can you try a number?" before giving any other hint.

Ask

"How did you start? What would you try next time?"

7 The sum of two consecutive numbers (numbers that are next to each other, like 40 and 41) is 145. What are the two numbers?

Go further

8 At the first bus stop, 1/3 of the passengers got off. At the second stop, half of those left got off and 3 got on. There are now 12 passengers. How many were on the bus at first?

9 How many rectangles of all sizes are there in a 1 by 5 strip of 5 small squares?

LOOK AFTER YOUR CHILD HAS TRIED

ALL LESSONS

Answers

Week 1 • Lesson 1: Million Puzzles

1. Four million, thirty thousand, two hundred and five.
2. 600 000
3. 5 483 000
4. Greatest 6 543 210; smallest 1 023 456.
5. 7 numbers: 1 000 001, 1 000 010, 1 000 100, 1 001 000, 1 010 000, 1 100 000 and 2 000 000.
6. (2 + 3) × 4 + 5 = 25 and 2 + 3 × (4 + 5) = 29. (29 is the biggest you can make with one pair.)
7. Range: 2 499 500 to 2 500 499. Six numbers: 2 500 002, 2 500 011, 2 500 020, 2 500 101, 2 500 110, 2 500 200. None below 2 500 000: 2 499 5__ already has digits adding to at least 29.
8. Open-ended. Check the riddle by listing every number that fits the rounding clue first. Sample riddle with exactly one answer (checked on a computer over all 7-digit numbers): all digits different; rounds to 5 400 000 to the nearest hundred thousand; the ten thousands digit is 1 less than the hundred thousands digit; the thousands digit is the sum of those two digits; the tens digit is 0 and the hundreds digit is even; the digits add up to 30. Answer: 5 437 209.
9. 64 × 15 625. (1 000 000 = 10 × 10 × 10 × 10 × 10 × 10; put all the 2s together to make 64 and all the 5s together to make 15 625.)

Week 1 • Lesson 2: Fraction Feats

1. 3/4 of a bar (cut every bar into 4 pieces: 12 pieces, 3 each; 3 ÷ 4 = 3/4).
2. 3/4 eaten, 1/4 left.
3. 4 + 3 = 7 slices eaten; 5 slices are left, which is 5/12 of the pizza.
4. 9 (each group has 3 dots, and 3 groups make 9).
5. 1/3 + 1/24 and 1/4 + 1/8. Test 1/3, 1/4, 1/5, ... in order: 1/5 leaves 7/40, not a unit fraction; after that the first fraction would be less than half of 3/8, so stop.
6. 1/2. 2/3 was left after Ben; Ali ate 1/4 of it, so 3/4 of 2/3 = 1/2 is left.
7. 1/20 and 1/90 (the bottom numbers multiply: 4 × 5 and 9 × 10).
8. Only one way: 1/2 + 1/3 + 1/6. (The biggest part must be 1/2, and then 1/2 = 1/3 + 1/6 is the only split into two different unit fractions.)
9. 1/100. Each top number cancels the bottom number before it, leaving 1 on top and 100 at the bottom.

Week 2 • Lesson 3: Before and After

1. Ali 14, Mei 10. The total is 24 before and 24 after.
2. Ben 15 and his father 43. The father is 28 years older now, and 28 years older in 5 years (the difference stays the same).
3. $30 (book $15, remainder $45, T-shirt $15)
4. 4 (the total is 60, so 30 each)
5. $120 (20 units: Monday 5 units, Tuesday 6 units; 11 units = $66, so 1 unit = $6)
6. 70 at first, 100 now (the blue beads did not change: 4 units at first, and 2 parts of 5 after, so 1 part = 2 units and the whole box is 10 units; red 3 units at first and 6 units now, so 3 units = 30 and 1 unit = 10)
7. 1 year ago (the difference, 32, is 4 units; 1 unit = 8: son 8, Mrs Tan 40)
8. Open-ended. Sample: Ali and Ben have 108 stickers altogether. Ali gives Ben 24 stickers. Then Ali has 1/2 as many stickers as Ben. How many did Ali have at first? (After: 3 units = 108, Ali 36, Ben 72; Ali had 36 + 24 = 60, Ben 48.) Check that the total is the same in both models and that every unit is the same size.
9. Eight ways (Ali, Mei): 18 and 6 (Ali gives none), 16 and 8, 8 and 16, 6 and 18, 4 and 20, 3 and 21, 2 and 22, 1 and 23. The total stays 24, so the smaller number must go into 24 exactly and be less than 12: 1, 2, 3, 4, 6 or 8. (12 and 12 is equal, so it does not count.)

Week 2 • Lesson 4: Triangle Areas

1. 12 cm^2 (the rectangle is 24 cm^2 and each triangle is half of it).
2. 32 cm^2. The two side pieces cover the middle triangle exactly, so the triangle is half of the square (64 ÷ 2).
3. 6 cm^2 (the rectangle is 4 × 3 = 12 cm^2 and one triangle is half of it).
4. 10 cm^2, and it does not change when the point moves along the top side (the triangle is always half of the rectangle).
5. 9 cm
6. 96 cm^2 (1/2 × 12 × 16)
7. 8 cm (the square is 36 cm^2, and 1/2 × 9 × 8 = 36).
8. Open-ended. Sample: ABCD is a rectangle 10 cm by 6 cm, E is the middle of the top side AB and F is the middle of the right side BC. The shaded triangle is DEF (D is the bottom-left corner). Taking away: 60 − 15 (ADE) − 7.5 (EBF) − 15 (FCD) = 22.5 cm^2. Check by a second method, for example by drawing the puzzle on 1-cm grid paper and counting squares, that the answers agree.
9. 36 cm^2 (96 − 24 − 12 − 24)

Week 3 • Lesson 5: Volume Vault

1. 6 cubes in the bottom layer, 2 layers, 12 cubes altogether (12 cm^3).
2. 3 rectangles: 1 by 12, 2 by 6 and 3 by 4.
3. 300 cm^3; 1000 cm^3, which is 1 ℓ.
4. 60 cubes (a layer of 20, three layers).
5. 24 (4 along, 3 across, 2 layers; not 270 ÷ 8, because the cubes cannot be cut)
6. 45 ℓ (3/5 − 1/3 = 4/15 of the tank is 12 ℓ, so 1/15 is 3 ℓ)
7. 7 cm (base area 245 ÷ 5 = 49 cm^2, and 7 × 7 = 49)
8. 216 (each face has 96 ÷ 6 = 16 = 4 × 4 one-face cubes, so the cube is 6 cubes along each edge: 6 × 6 × 6)
9. 7 (1 × 1 × 64, 1 × 2 × 32, 1 × 4 × 16, 1 × 8 × 8, 2 × 2 × 16, 2 × 4 × 8, 4 × 4 × 4)

Week 3 • Lesson 6: Rate Racers

1. $4; $6; 50¢ each.
2. 15 minutes
3. Pack A. Two Pack A cost $4 for 6 pens, but Pack B costs $5 for 6 pens. (Pack A is about 67¢ a pen and Pack B is about 83¢ a pen.)
4. 15 minutes
5. 5 ℓ a minute (the tank holds 180 ℓ; 180 ÷ 9 = 20 ℓ a minute together; 20 – 15 = 5)
6. $9.60 (one box of 10 and three boxes of 6 make 28 eggs; three boxes of 10 cost $9.90)
7. $9.50 (4 h 35 min: first hour, then 3 h 35 min is 8 half hours or parts)
8. 2 minutes (pretend the job is 6 pages: P prints 1 page a minute, Q 2 pages, together 3 a minute)
9. 30 pens (6 pens cost $4 and sell for $6: $2 profit for every 6 pens)

Week 4 • Lesson 7: Percentage Puzzles

1. 35% shaded, 65% not shaded.
2. 50% is $40, 25% is $20 and 10% is $8.
3. $5 is taken off; you pay $45.
4. 60% (24 out of 40 = 60 out of 100)
5. (a) 25% ($1 out of $4) (b) 20% ($1 out of $5) (c) Each change is compared with its own starting amount: $1 out of $4 is a bigger part than $1 out of $5. (This is a puzzle that goes further than the lesson.)
6. (a) Shop X $45 (she pays for 3 boxes); Shop Y $42 (30% of $60 = $18 off). Shop Y is the better buy, by $3. (b) 25% off (1 box free out of 4, $15 off $60)
7. She found 10% ($8) and 5% (half of 10%, $4) and added them. 15% of $46 = $4.60 + $2.30 = $6.90.
8. Open-ended. Sample: a $60 jacket goes up 50% and then down 50%. Ravi says the price is back to $60. Really: $60 + $30 = $90, then 50% of $90 is $45 off, so $45. It is 25% LESS than the start, because the second 50% is of the bigger amount.
9. No. $40 is 80% of the old price, so 10% is $5 and 100% is $50. (Ben added 20% of $40, but the 20% was of the old price.)

Week 4 • Lesson 8: Angle Chasers

1. The three corners fit together along a straight line, so the three angles add up to 180°.
2. 53° (180° – 127°).
3. 45° for one slice (360° ÷ 8) and 90° for two slices.
4. About 65° (180° – 48° – 67°). The three angles add up to 180°.
5. A = 135° and B = 45°. Adjacent angles of a parallelogram add up to 180°, so A and B make 4 equal parts of 45°: B = 45°, A = 135°.
6. 35° and 35°. 110° cannot be a base angle (two of them would be 220°), so it is the top angle and 180° – 110° = 70° is shared by two equal base angles.
7. AEC = 118° and EAB = 62°. ADC = 56° (opposite angles); DEA = (180° – 56°) ÷ 2 = 62° (DA = DE); AEC = 180° – 62° = 118° (straight line DC); DAB = 180° – 56° = 124° (neighbours in a parallelogram), so EAB = 124° – 62° = 62°.
8. 30°, 30°, 120°. The two right angles make a straight line, and the two 60° angles together make 120°.
9. Open-ended. Sample: ABCD is a parallelogram with angle ABC = 56°, and E is on DC with DA = DE. Find angle EAB. (ADC = 56°, DEA = (180° – 56°) ÷ 2 = 62°, DAB = 180° – 56° = 124°, EAB = 124° – 62° = 62°.) Check that every step has a reason and that the given angles do not contradict each other.

Week 5 • Lesson 9: Pattern Generalisers

1. 41 (5 + 9 × 4: there are 9 jumps of 4 from the 1st number to the 10th).
2. 25 (3 × 8 + 1).
3. Yes. 50 − 2 = 48 and 48 ÷ 4 = 12 jumps, so 50 is the 13th number.
4. 7 tricycles (suppose all 15 are bicycles: 30 wheels; 7 wheels short; each tricycle adds 1 more wheel). 8 bicycles. Check: 16 + 21 = 37.
5. Red (50 = 8 × 6 + 2, so it is the 2nd bead of a group). 49 yellow beads: 16 full groups of 6 give 96 beads and 48 yellow, and beads 97 to 100 are red, red, blue, yellow.
6. 96 (each row of 5 seats 12, and 8 × 12 = 96). One long row seats only 82. Splitting the tables gives more end seats.
7. 12 goats and 23 chickens (all chickens would have 70 legs; 24 legs short; each goat adds 2 legs). Check: 48 + 46 = 94.
8. 55 (1 + 2 + 3 + … + 10). Two copies of the staircase make a 10 by 11 rectangle: 110 ÷ 2 = 55.
9. Open-ended. Sample: matchstick triangles in a row use 3, 5, 7, … sticks, so sticks = 2 × figure number + 1. Figure 50 uses 101 sticks. No figure uses exactly 100 sticks, because 2 × figure number + 1 is always odd (figure 49 uses 99 and figure 50 uses 101). Check that the rule works for figures 1, 2 and 3.

Week 5 • Lesson 10: Challenge Day

1. 14 (work backwards: 50 + 6 = 56, 56 ÷ 4 = 14).
2. $18 ($9 is the other half).
3. 24 and 16 (take away the 8: 32, share: 16 and 16, then add 8 back to one: 24 and 16).
4. Siti. Ravi has the rabbit (no bird, no hamster). Siti is not the hamster, so she has the bird, and Ben has the hamster.
5. $60. Work backwards: $24 is 2/3 of what was left, so that was $36; $36 is 3/5 of her money, so 1/5 is $12 and she started with $60. Check: 60 − 24 = 36, 36 − 12 = 24.
6. 15 numbers: 104, 113, 122, 131, 140, 203, 212, 221, 230, 302, 311, 320, 401, 410, 500.
7. 72 and 73 (145 is 1 more than twice 72).
8. 27. Work backwards: 12 − 3 = 9; 9 is half of what was left, so 18 were left after stop 1; 18 is 2/3 of the passengers, so 1/3 is 9 and there were 27 at first.
9. 15 (5 of length 1, 4 of length 2, 3 of length 3, 2 of length 4 and 1 of length 5: 5 + 4 + 3 + 2 + 1).